Supplementary Material for:

# Modal Analysis of a Fluid-Filled Viscoelastic Oblate Spheroidal Shell: Implications for the Existence of the Brown Note

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# 1 Detailed Derivations

This section provides the full derivation of all analytical results used in the main text. Equation numbering follows the convention (S$n$) to distinguish supplementary equations from those in the main paper.

## 1.1 Shell Flexural Rigidity

The abdominal wall is modelled as a thin isotropic shell of thickness $h$, Young's modulus $E$, and Poisson's ratio $\nu$. From classical Love–Kirchhoff thin-shell theory, the flexural (bending) rigidity is

$$D = \frac{E\,h^3}{12\,(1-\nu^2)}\,. \tag{S1}$$

For the canonical parameter set ($E = 0.1\,\mathrm{MPa}$, $h = 10\,\mathrm{mm}$, $\nu = 0.45$):

$$D = \frac{0.1\times 10^6\ \times\ (0.01)^3}{12\,(1-0.45^2)} = \frac{10^5\times 10^{-6}}{12\times 0.7975} \approx 0.0104\,\mathrm{N\,m}\,.$$

The thin-shell condition $h/R \approx 0.06 \ll 1$ is satisfied for the equivalent-sphere radius $R_{\mathrm{eq}} = (a^2c)^{1/3} \approx 0.157\,\mathrm{m}$.

## 1.2 Added Mass Coefficient for Mode $n$ (Lamb's Result)

For flexural mode $n$ ($n \geq 2$) of a thin spherical shell of radius $R$ filled with an incompressible fluid of density $\rho_f$, the interior velocity potential is expanded in Legendre polynomials. The solution to Laplace's equation inside a sphere for mode $n$ gives a radial dependence $\phi \propto r^n$, so the fluid velocity at the shell wall scales as $r^{n-1}$.

Integrating the kinetic energy of the internal flow, Lamb (1882) showed that the fluid added mass per unit shell area is

$$m_{\mathrm{add}}^{(n)} = \frac{\rho_f\,R}{n}\,. \tag{S2}$$

This result has a transparent physical interpretation: for higher mode numbers $n$, the fluid motion is more localised near the wall (shorter "penetration depth" $\sim R/n$), entraining less fluid mass.

For the $n = 2$ oblate–prolate mode with $\rho_f = 1020\,\mathrm{kg/m^3}$ and $R = 0.157\,\mathrm{m}$:

$$m_{\mathrm{add}}^{(2)} = \frac{1020\times 0.157}{2} = 80.1\,\mathrm{kg/m^2}\,.$$

Compare this with the shell wall mass per unit area, $m_{\mathrm{wall}} = \rho_w\,h = 1100\times 0.01 = 11.0\,\mathrm{kg/m^2}$. The fluid added mass exceeds the wall mass by a factor of $\sim$7.3, confirming that fluid inertia dominates the effective modal mass.

## 1.3 Flexural Frequency Formula with Intermediate Steps

The flexural modes ($n \geq 2$) involve shell deformation at approximately constant enclosed volume; the fluid acts as added mass but does not contribute volumetric (bulk) stiffness. The restoring force per unit shell area comprises three terms:

**Bending stiffness.** From Love–Kirchhoff theory for a spherical shell:

$$K_{\text{bend}}^{(n)} = \frac{n\,(n-1)\,(n+2)^2\,D}{R^4}\,. \tag{S3}$$

**Membrane stiffness.** Lamb (1882) obtained the membrane contribution:

$$K_{\text{memb}}^{(n)} = \frac{E\,h}{R^2}\,\lambda_n\,, \qquad \lambda_n = \frac{n^2+n-2+2\nu}{n^2+n+1-\nu}\,. \tag{S4}$$

**Pre-stress stiffness.** The intra-abdominal pressure $P_{\text{iap}}$ prestresses the shell in biaxial tension, adding a geometric stiffness:

$$K_P^{(n)} = \frac{P_{\text{iap}}}{R}\,(n-1)(n+2)\,. \tag{S5}$$

**Total stiffness and effective mass.** The total stiffness per unit area is

$$K_{\text{total}}^{(n)} = K_{\text{bend}}^{(n)} + K_{\text{memb}}^{(n)} + K_P^{(n)}\,, \tag{S6}$$

and the effective mass per unit area is

$$m_{\text{eff}}^{(n)} = \rho_w\,h + \frac{\rho_f\,R}{n}\,. \tag{S7}$$

The natural frequency follows as

$$\boxed{f_n = \frac{1}{2\pi}\sqrt{\frac{K_{\text{total}}^{(n)}}{m_{\text{eff}}^{(n)}}}}\,. \tag{S8}$$

**Numerical evaluation for $n = 2$.** With the canonical parameters ($E = 0.1\,\text{MPa}$, $a = 0.18\,\text{m}$, $c/a = 0.67$, $h = 10\,\text{mm}$, $\nu = 0.45$, $\rho_w = 1100\,\text{kg/m}^3$, $\rho_f = 1020\,\text{kg/m}^3$, $P_{\text{iap}} =$

1000 Pa):

$$
\begin{aligned}
R_{\text{eq}} &= (0.18^2 \times 0.12)^{1/3} = 0.157\,\text{m}, \\
D &= 0.0104\,\text{N}\,\text{m}, \\
K_{\text{bend}}^{(2)} &= \frac{2 \times 1 \times 16 \times 0.0104}{0.157^4} = 547\,\text{Pa/m}, \\
\lambda_2 &= \frac{4+2-2+0.90}{4+2+1-0.45} = \frac{4.90}{6.55} = 0.748, \\
K_{\text{memb}}^{(2)} &= \frac{10^5 \times 0.01}{0.157^2} \times 0.748 = 30\,363\,\text{Pa/m}, \\
K_P^{(2)} &= \frac{1000}{0.157} \times 1 \times 4 = 25\,478\,\text{Pa/m}, \\
K_{\text{total}}^{(2)} &= 547 + 30\,363 + 25\,478 = 56\,388\,\text{Pa/m}, \\
m_{\text{eff}}^{(2)} &= 1100 \times 0.01 + \frac{1020 \times 0.157}{2} = 11.0 + 80.1 = 91.1\,\text{kg/m}^2, \\
f_2 &= \frac{1}{2\pi}\sqrt{\frac{56\,388}{91.1}} \approx 4.0\,\text{Hz}\,.
\end{aligned}
$$

## 1.4 Breathing Mode Derivation

The breathing mode ($n = 0$) involves a uniform radial expansion/contraction of the shell. Unlike flexural modes, this motion *does* compress the enclosed fluid, so the fluid bulk modulus $K_f$ contributes directly to the stiffness.

**Shell membrane stiffness.** For a uniform radial displacement $\delta R$, the circumferential strain is $\varepsilon = \delta R/R$. Biaxial membrane stress gives a restoring pressure:

$$k_{\text{shell}} = \frac{2\,E\,h}{R^2\,(1-\nu)}\,. \tag{S9}$$

**Fluid volumetric stiffness.** The volume change is $\Delta V = 4\pi R^2\,\delta R$, and the pressure change is $\Delta p = K_f\,\Delta V/V = 3\,K_f\,\delta R/R$. The restoring force per unit shell area is therefore

$$k_{\text{fluid}} = \frac{3\,K_f}{R}\,. \tag{S10}$$

**Comparison of stiffness contributions.**

$$
\begin{aligned}
k_{\text{shell}} &= \frac{2 \times 10^5 \times 0.01}{0.157^2 \times 0.55} \approx 1.5 \times 10^5\,\text{Pa/m}, \\
k_{\text{fluid}} &= \frac{3 \times 2.2 \times 10^9}{0.157} \approx 4.2 \times 10^{10}\,\text{Pa/m}.
\end{aligned}
$$

The fluid volumetric stiffness exceeds the shell membrane stiffness by a factor of $\sim 3 \times 10^5$; hence $K_f$ completely dominates the breathing mode stiffness.

**Effective mass and frequency.**

$$m_{\text{eff}}^{(0)} = \rho_w\, h + \rho_f\, R = 11.0 + 160.1 = 171.1\,\text{kg/m}^2, \tag{S11}$$

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{k_{\text{shell}} + k_{\text{fluid}}}{m_{\text{eff}}^{(0)}}} \approx \frac{1}{2\pi}\sqrt{\frac{4.2\times 10^{10}}{171.1}} \approx 2500\,\text{Hz}\,. \tag{S12}$$

This is far above the infrasonic range ($< 20\,\text{Hz}$) and is therefore irrelevant to the brown-note hypothesis.

## 1.5 Energy-Consistent Displacement Derivation

The pressure-based estimate of airborne coupling displacement (main text, Section 4) is approximately 13× larger than the energy-consistent value because it does not enforce energy conservation. The self-consistent derivation, following the reciprocity formulation of Junger & Feit [**?**], proceeds as follows.

**Step 1: Absorption cross-section.** By reciprocity, the maximum absorption cross-section at resonance for a mode of order $n$ is

$$\sigma_{\text{max}}^{(n)} = \frac{(2n+1)\,\lambda^2}{4\pi}\,, \tag{S13}$$

where $\lambda = c_{\text{air}}/f_n$ is the acoustic wavelength. The actual cross-section is limited by the ratio of radiation damping to total damping:

$$\sigma_{\text{abs}}^{(n)} = \sigma_{\text{max}}^{(n)} \times \frac{\zeta_{\text{rad}}}{\zeta_{\text{rad}} + \zeta_{\text{struct}}}\,. \tag{S14}$$

**Step 2: Radiation damping.** The radiation resistance for mode $n$ of a sphere in the Rayleigh limit ($ka \ll 1$) is

$$R_{\text{rad}}^{(n)} = \rho_{\text{air}}\, c_{\text{air}}\, 4\pi R^2 \times \frac{(ka)^{2n+2}}{[(2n+1)!!]^2}\,, \tag{S15}$$

where $(2n+1)!! = 1\times 3\times 5\times\cdots\times(2n+1)$ is the double factorial. For $n = 2$ at $f = 4\,\text{Hz}$:

$$\begin{gathered} ka = \frac{2\pi\times 4\times 0.157}{343} \approx 0.012, \\ (ka)^6 \approx 2.4\times 10^{-12}, \qquad 5!! = 15, \\ \zeta_{\text{rad}} \sim 10^{-15} \ll \zeta_{\text{struct}} = 0.125\,. \end{gathered}$$

**Step 3: Steady-state energy balance.** At steady state, absorbed power equals dissipated power:

$$\sigma_{\text{abs}}\, I = \zeta_{\text{struct}}\, \omega_n^3\, M_{\text{total}}\, \xi^2\,, \tag{S16}$$

where $I = p_{\text{inc}}^2/(2\rho_{\text{air}} c_{\text{air}})$ is the incident intensity, $M_{\text{total}} = m_{\text{eff}} \times 4\pi R^2$ is the total modal mass, and $\xi$ is the displacement amplitude.

Solving for the energy-consistent displacement:

$$\boxed{\xi_{\text{energy}} = \sqrt{\frac{\sigma_{\text{abs}}\, I}{\zeta_{\text{struct}}\, \omega_n^3\, M_{\text{total}}}}}. \tag{S17}$$

**Comparison with the pressure-based estimate.** At 120 dB SPL and $n = 2$:

$$\xi_{\text{energy}} \approx 0.014\,\mu\text{m},$$
$$\xi_{\text{pressure}} \approx 0.18\,\mu\text{m}.$$

The pressure-based estimate is approximately 13× larger because it does not account for the extremely low radiation efficiency ($\zeta_{\text{rad}}/\zeta_{\text{struct}} \sim 10^{-14}$). Both values are far below the PIEZO1 and PIEZO2 mechanotransduction threshold range (0.5 µm to 2.0 µm), so the overestimate does not affect any conclusions.

# 2 Complete Parameter Tables

## 2.1 Canonical Parameter Set

Table 1 lists the baseline parameter values used throughout the analysis, together with literature sources.

Table 1: Canonical (baseline) parameter set.

| Parameter | Symbol | Value | Unit | Source |
|---|---|---|---|---|
| Semi-major axis | $a$ | 0.18 | m | Anthropometric mean |
| Semi-minor axis | $c$ | 0.12 | m | $c/a \approx 0.67$ |
| Wall thickness | $h$ | 0.010 | m | Composite wall |
| Young's modulus | $E$ | 0.1 | MPa | Relaxed passive tissue; Fung (1993) |
| Poisson's ratio | $\nu$ | 0.45 | — | Nearly incompressible |
| Wall density | $\rho_w$ | 1100 | kg/m$^3$ | Soft tissue |
| Fluid density | $\rho_f$ | 1020 | kg/m$^3$ | Abdominal fluid |
| Fluid bulk modulus | $K_f$ | 2.2 | GPa | Water at 37 °C |
| Intra-abdominal pressure | $P_{\text{iap}}$ | 1000 | Pa | Resting supine; ∼7.5 mmHg |
| Loss tangent | $\eta$ | 0.25 | — | Viscoelastic tissue |
| *Derived quantities* | | | | |
| Equivalent sphere radius | $R_{\text{eq}}$ | 0.157 | m | $(a^2c)^{1/3}$ |
| Flexural rigidity | $D$ | 0.0104 | N m | Eq. (S1) |
| Damping ratio | $\zeta$ | 0.125 | — | $\eta/2$ |
| Quality factor | $Q$ | 4.0 | — | $1/(2\zeta)$ |

## 2.2 Monte Carlo Parameter Distributions

Table 2 specifies the probability distributions used in the uncertainty quantification ($N$ = 10,000 Monte Carlo samples, Sobol base $N_{\text{Sobol}} = 4096$, seed = 42).

Table 2: Parameter distributions for Monte Carlo / Sobol analysis.

| Parameter | Nominal | Distribution | Low | High |
|---|---|---|---|---|
| $E$ (Pa) | $10^5$ | Lognormal | $5 \times 10^4$ | $2 \times 10^6$ |
| $a$ (m) | 0.18 | Normal | 0.15 | 0.22 |
| $c$ (m) | 0.12 | Normal | 0.08 | 0.15 |
| $h$ (m) | 0.01 | Normal | 0.005 | 0.02 |
| $\nu$ | 0.45 | Uniform | 0.40 | 0.499 |
| $\rho_w$ (kg/m$^3$) | 1100 | Normal | 1000 | 1200 |
| $\rho_f$ (kg/m$^3$) | 1020 | Normal | 1000 | 1060 |
| $P_{\text{iap}}$ (Pa) | 1000 | Lognormal | 500 | 3000 |
| $\eta$ | 0.25 | Uniform | 0.10 | 0.40 |

## 2.3 Complete Sobol Sensitivity Indices

Tables 3–6 present the first-order ($S_1$) and total-order ($S_T$) Sobol indices for each output quantity.

Table 3: Sobol indices for $n = 2$ flexural frequency $f_2$.

| Parameter | $S_1$ | $S_T$ |
|---|---|---|
| $E$ | 0.8119 | 0.8613 |
| $a$ | 0.0353 | 0.0503 |
| $c$ | 0.0272 | 0.0359 |
| $h$ | 0.0659 | 0.1163 |
| $\nu$ | 0.000 13 | 0.000 24 |
| $\rho_w$ | 0.000 079 | 0.000 093 |
| $\rho_f$ | 0.000 033 | 0.000 24 |
| $P_{\text{iap}}$ | 0.0108 | 0.0125 |
| $\eta$ | 0.0000 | 0.0000 |

## 2.4 Elastic Modulus Sweep

Table 7 shows the sensitivity of the fundamental flexural frequency to the elastic modulus, the parameter that dominates modal uncertainty ($S_T = 0.86$).

Table 4: Sobol first-order indices for airborne displacement $\xi_{\text{air}}$ at 120 dB.

| Parameter | $S_1$ |
|---|---|
| $E$ | $< 0.001$ |
| $a$ | 0.041 |
| $c$ | 0.028 |
| $h$ | 0.011 |
| $\nu$ | $< 0.001$ |
| $\rho_w$ | $< 0.001$ |
| $\rho_f$ | 0.001 |
| $P_{\text{iap}}$ | $< 0.001$ |
| $\eta$ | 0.899 |

Table 5: Sobol indices for mechanical displacement $\xi_{\text{mech}}$ at $0.5\,\text{m/s}^2$.

| Parameter | $S_1$ |
|---|---|
| $E$ | 0.244 |
| $a$ | 0.015 |
| $c$ | 0.014 |
| $h$ | 0.102 |
| $\nu$ | $< 0.001$ |
| $\rho_w$ | $< 0.001$ |
| $\rho_f$ | $< 0.001$ |
| $P_{\text{iap}}$ | 0.022 |
| $\eta$ | 0.089 |

Table 6: Sobol indices for coupling ratio $\mathcal{R} = \xi_{\text{mech}}/\xi_{\text{air}}$.

| Parameter | $S_1$ |
|---|---|
| $E$ | 0.331 |
| $a$ | 0.005 |
| $c$ | 0.006 |
| $h$ | 0.088 |
| $\nu$ | $< 0.001$ |
| $\rho_w$ | $< 0.001$ |
| $\rho_f$ | 0.001 |
| $P_{\text{iap}}$ | 0.027 |
| $\eta$ | $< 0.001$ |

Table 7: $n = 2$ frequency vs. elastic modulus.

| $E$ (MPa) | $f_2$ (Hz) | ISO 2631 (4–8 Hz)? |
|---|---|---|
| 0.05 | 3.4 | Marginal |
| 0.10 | 4.0 | Yes |
| 0.20 | 4.9 | Yes |
| 0.50 | 7.1 | Yes |
| 1.00 | 9.6 | Marginal |
| 2.00 | 13.3 | No |

## 2.5 Parametric Study Overview

A full factorial parametric study explored 486 combinations: 6 values of $E$ × 3 semi-major axes × 3 aspect ratios × 3 wall thicknesses × 3 loss tangents.

Key findings from the 486-combination sweep:

- 37% of configurations place $f_2$ in the 5–10 Hz range.
- 41% of configurations place $f_2$ in the ISO 2631 range (4–8 Hz).
- Dominant factor: $E$, followed by geometry ($a$, $c/a$) and $P_{\text{iap}}$.

Table 8: Representative body-type configurations and ISO 2631 overlap.

| Body type | $E$ (MPa) | $a$ (cm) | $f_2$ (Hz) | In ISO range? |
|---|---|---|---|---|
| Lean | 0.2 | 15 | 6.3 | Yes |
| Average | 0.1 | 18 | 4.0 | Yes |
| Large | 0.08 | 20 | 3.1 | Marginal |

## 2.6 Monte Carlo Summary Statistics

Table 9 summarises the full Monte Carlo output distributions ($N = 10{,}000$ samples).

Table 9: Monte Carlo summary statistics ($N = 10{,}000$).

| Quantity | Mean | Median | Std Dev | P5 | P95 | 99% CI |
|---|---|---|---|---|---|---|
| $f_2$ (Hz) | 7.52 | 6.43 | 4.21 | 3.32 | 15.48 | 26. |
| $\xi_{\text{air}}$ (µm) | 0.206 | 0.178 | 0.090 | 0.107 | 0.394 | 0. |
| $\xi_{\text{mech}}$ (µm) | 2745 | 1818 | 3110 | 296 | 8176 | 17 457 |
| $\mathcal{R}$ | 12 949 | 9820 | 11 888 | 1780 | 34 169 | 62 466 |
| $\text{SPL}_{\text{PIEZO}}$ (pressure)[1] (dB) | 134.4 | 135.0 | 3.5 | 128.1 | 139.4 | 141. |

# 3 Sensitivity to Modelling Assumptions

This section summarises validation studies that quantify the error introduced by each simplifying assumption in the analytical model.

## 3.1 Oblate Spheroid vs. Equivalent Sphere

The main-text model uses an equivalent sphere of radius $R_{\text{eq}} = (a^2c)^{1/3}$. A Rayleigh–Ritz analysis on the full oblate spheroid (Section 1) was performed using `oblate_spheroid_ritz.py` with a 2-DOF trial function per mode (normal and tangential displacements expressed as Legendre polynomials), integrated over the oblate surface via 200-point Gauss–Legendre quadrature.

Table 10: Sphere vs. oblate Rayleigh–Ritz frequencies at $c/a = 0.67$.

| | $E = 0.1\,\text{MPa}$ | | $E = 0.5\,\text{MPa}$ | |
|---|---|---|---|---|
| Mode | Sphere (Hz) | Ritz (Hz) | Sphere (Hz) | Ritz (Hz) |
| $n = 2$ | 4.0 | 3.8 | 7.1 | 6.2 |
| $n = 3$ | 6.3 | 5.8 | 10.0 | 8.5 |
| $n = 4$ | 8.9 | 8.1 | 12.9 | 11.3 |

**Findings.** The sphere model overestimates flexural frequencies by 4–17%. The error grows with eccentricity and with $n$. Physical reasons: (i) spatially varying curvature on the oblate surface (poles sharper, equator flatter), (ii) different fluid added-mass distribution from oblate flow geometry, and (iii) the Ritz formulation captures optimal tangential displacement via the variational principle. Despite the systematic offset, the conclusion that $f_2$ lies in the 3 Hz to 8 Hz range is robust to the choice of geometry model.

## 3.2 Multilayer vs. Homogeneous Wall

The real abdominal wall comprises ∼5–6 distinct layers. Table 11 shows the layer stack used in `multilayer_wall.py`.

Effective composite properties are computed using classical laminate theory:

$$(Eh)_{\text{eff}} = \sum_i E_i\, h_i\,, \tag{S18}$$

$$D_{\text{eff}} = \sum_i \frac{E_i}{1-\nu_i^2}\left[\frac{h_i^3}{12} + h_i(z_i - \bar{z})^2\right], \tag{S19}$$

$$\rho_{\text{eff}} = \frac{\sum_i \rho_i\, h_i}{\sum_i h_i}\,. \tag{S20}$$

Table 11: Relaxed abdominal wall layer properties.

| Layer | $h$ (mm) | $E$ (MPa) | $\nu$ | $\rho$ (kg/m$^3$) |
|---|---|---|---|---|
| Peritoneum | 0.1 | 0.50 | 0.45 | 1050 |
| Transversalis | 3.0 | 0.05 | 0.45 | 1050 |
| Internal oblique | 5.0 | 0.05 | 0.45 | 1060 |
| External oblique | 5.0 | 0.08 | 0.45 | 1060 |
| Subcutaneous fat | 10.0 | 0.002 | 0.49 | 950 |
| Skin | 2.0 | 0.50 | 0.45 | 1100 |
| **Composite** | **25.1** | **0.075** | **0.473** | **1012** |

**Key result.** The multilayer relaxed model predicts $f_2 \approx 4.6\,\mathrm{Hz}$, within approximately 16% of the homogeneous model at $E = 0.1\,\mathrm{MPa}$. The tensed muscle configuration ($E_{\mathrm{eff}} \approx 2.7\,\mathrm{MPa}$) raises $f_2$ to $\sim$23 Hz, confirming that the dominant variable is muscle tone, not the specific layer arrangement.

## 3.3 Viscous vs. Inviscid Fluid

Fluid viscosity introduces a Stokes oscillatory boundary layer of thickness

$$\delta = \sqrt{\frac{2\mu}{\rho_f\,\omega}} \tag{S21}$$

at the inner wall. The associated viscous damping ratio addition is obtained from a standard oscillatory boundary-layer calculation:

$$\Delta\zeta_{\mathrm{visc}} = \frac{\rho_f\,\delta\,(n+1)}{4\,n\,(2n+1)\,m_{\mathrm{eff}}^{(n)}}\,. \tag{S22}$$

Table 12: Viscous correction for physiological fluids ($n = 2$, $f = 4\,\mathrm{Hz}$).

| Fluid | $\mu$ (Pa s) | $\delta$ (mm) | $\Delta\zeta_{\mathrm{visc}}$ | $\Delta\zeta/\zeta_{\mathrm{struct}}$ (%) | $\Delta Q$ (%) |
|---|---|---|---|---|---|
| Water | 0.001 | $\sim$0.2 | $\sim 10^{-4}$ | $< 0.1$ | $< 0.1$ |
| Bile | 0.005 | $\sim$0.4 | $\sim 3\times10^{-4}$ | $< 0.3$ | $< 0.3$ |
| Blood | 0.004 | $\sim$0.4 | $\sim 2\times10^{-4}$ | $< 0.2$ | $< 0.2$ |
| Gastric acid | 0.01 | $\sim$0.6 | $\sim 4\times10^{-4}$ | $< 0.4$ | $< 0.4$ |
| Mucus | 0.10 | $\sim$2.0 | $\sim 10^{-3}$ | $< 1.0$ | $< 1.0$ |

**Key result.** Even for the most viscous physiological fluid (mucus, $\mu = 0.1\,\mathrm{Pa\,s}$), the viscous damping adds $< 1\%$ to the structural damping. The crossover viscosity at which $\Delta\zeta_{\mathrm{visc}} = \zeta_{\mathrm{struct}}$ is $\mu_{\mathrm{cross}} \approx 280\,\mathrm{Pa\,s}$—comparable to honey or peanut butter and far beyond any abdominal fluid. The inviscid approximation is therefore well justified.

**Non-Newtonian effects.** Chyme is shear-thinning (power-law exponent $n_{\rm pl} \approx 0.5$–$0.8$). At the extremely low shear rates characteristic of infrasonic oscillation ($\dot{\gamma} \sim 0.1/\mathrm{s}$), the apparent viscosity is *higher* than the consistency index, but the viscous damping correction remains negligible ($\Delta\zeta/\zeta_{\rm struct} < 2\%$).

## 3.4 With and Without Organ Inclusions

Solid organs (liver, spleen, kidneys) occupy ∼20–30% of the cavity volume with shear moduli of 1–10 kPa.

The analysis (`organ_inclusions.py`) uses Hashin–Shtrikman bounds for the effective medium. Key results:

1. **Bulk modulus**: Since both fluid and tissue have $K \approx 2.2\,\mathrm{GPa}$ (nearly incompressible), the effective bulk modulus changes by $< 1\,\mathrm{ppm}$—negligible.
2. **Shear modulus**: The HS lower bound gives $G_{\rm eff} = 0$ for the composite because the fluid matrix cannot transmit shear at the macroscale (inclusions are non-percolating, $\phi < 0.29$).
3. **Density**: The effective density is a simple volume average; organ tissue ($\rho \approx 1055\,\mathrm{kg/m^3}$) vs. fluid ($\rho = 1020\,\mathrm{kg/m^3}$) gives $\Delta\rho/\rho \approx 1\%$.
4. **Frequency shift**: The combined stiffness and mass perturbation shifts $f_2$ by <1%. The liver's asymmetric mass (1.5 kg) lifts the $m$-degeneracy of the $n = 2$ mode by ∼1–2%.

**Conclusion.** Treating the cavity contents as homogeneous fluid introduces $< 2\%$ frequency error, well within the uncertainty from the elastic modulus ($\sigma_E/E \sim$ factor of 5).

## 3.5 Nonlinear vs. Linear Analysis

For whole-body vibration at occupational levels, the predicted displacements ($\xi \sim 650$–$7500\,\mu\mathrm{m}$) approach $w/h \sim O(1)$, suggesting geometric nonlinearity may be relevant. The analysis (`nonlinear_analysis.py`) constructs a Duffing oscillator model from Donnell–von Kármán nonlinear shell theory.

**Cubic stiffness coefficient.** The von Kármán mid-surface strains introduce a cubic restoring force with coefficient

$$\alpha = \alpha_{\rm shell} + \alpha_{\rm fluid}\,, \tag{S23}$$

where

$$\alpha_{\rm shell} = \frac{E\,h}{4\,R^4}\,n^2(n+1)^2\,C_{\rm NL}\,, \quad C_{\rm NL} = \frac{3-\nu^2}{1-\nu^2}\,, \tag{S24}$$

$$\alpha_{\rm fluid} = \frac{3\,\rho_f\,\omega_0^2}{2\,R^2\,n}\,. \tag{S25}$$

The system is *hardening* ($\alpha > 0$): frequency increases with amplitude. For the $n = 2$ mode, $\alpha \approx 7.1 \times 10^7\,\mathrm{rad^2 s^{-2} m^{-2}}$.

**Backbone curve.** The amplitude-dependent resonance frequency is

$$\omega(A) = \omega_0 \sqrt{1 + \frac{3\,\alpha\,A^2}{4\,\omega_0^2}}\,. \tag{S26}$$

A 5% frequency shift occurs at a critical amplitude of $A_{\text{crit}} \approx 1092\,\mu\text{m}$ ($A/h \approx 0.11$).

**Amplitude-dependent damping.** Soft tissue exhibits increased dissipation at large strains, modelled as

$$\eta(A) = \eta_0\big(1 + \beta|A/h|^2\big)\,, \quad \beta = 0.5\,. \tag{S27}$$

This partially offsets the nonlinear stiffening by reducing peak amplitudes.

**Key result.** For airborne coupling ($\xi < 0.2\,\mu\text{m}$), nonlinearity is completely negligible ($A/h \sim 10^{-5}$). For mechanical coupling at occupational WBV levels, the nonlinear frequency shift is 5–15%, which stiffens the response and slightly reduces peak displacement but does not change the order-of-magnitude conclusion ($\xi_{\text{mech}} \gg$ PIEZO threshold).

## 3.6 Boundary Condition Effects

The Rayleigh–Ritz framework also investigated boundary condition sensitivity via FEA validation results (`fea_modal_results.json`). To match the mesh geometry, this study used a slightly different parameter set ($R = 0.162\,\text{m}$, $\rho_f = 1000\,\text{kg/m}^3$, $P_{\text{iap}} = 1500\,\text{Pa}$); the relative shifts between boundary conditions are therefore the parameter of interest.

Table 13: Boundary condition effects on $n = 2$ frequency (wet shell).

| Configuration | $f_2$ (Hz) | Shift from free (%) |
|---|---|---|
| Free sphere (Ritz) | 4.13 | 0 |
| Hemisphere simply-supported | 3.91 | −5.3 |
| Hemisphere clamped | 3.89 | −5.8 |
| 25% surface constrained | 3.66 | −11.4 |

**Key result.** The free-sphere Ritz value $f_2 = 4.13\,\text{Hz}$ agrees with the matched analytical prediction (4.26 Hz) to within 3%. The maximum frequency shift from boundary conditions is 11%. This insensitivity arises because the fluid added mass dominates ($m_{\text{add}}/m_{\text{wall}} \approx 7.3$), so the system dynamics are controlled by the fluid inertia rather than the shell boundary.

## 3.7 Summary of Modelling Assumptions

Table 14 compiles the frequency error from each modelling simplification.

Table 14: Impact of modelling assumptions on predicted $f_2$.

| Assumption | Error | Dominant factor |
|---|---|---|
| Sphere → oblate spheroid | 4–17% | Curvature distribution |
| Homogeneous → multilayer | ∼16% | Muscle tone |
| Inviscid → viscous fluid | <1% | Stokes boundary layer |
| No organs → with inclusions | <2% | Density perturbation |
| Linear → nonlinear | 5–15%* | WBV amplitudes only |
| Free → constrained BC | <11% | Fluid dominates |

*Only relevant for mechanical coupling ($\xi > 1\,\mathrm{mm}$).

All errors are small compared to the uncertainty in $E$ ($S_T = 0.86$; factor-of-5 range).

# 4 Code Availability

## 4.1 Repository

All source code, data, and scripts required to reproduce the results presented in the main paper and this supplementary material are available at:

https://github.com/JonathanMace/brownnote

## 4.2 Module Listing

Table 15 describes each module in the repository.

Table 15: Repository module listing.

| **Module / Script** | **Description** |
|---|---|
| *Analytical models* (`src/analytical/`) | |
| `natural_frequency_v2.py` | Flexural and breathing mode frequencies for a fluid-filled sphere with added mass, pre-stress, and viscoelastic damping. |
| `energy_budget.py` | Self-consistent energy budget via reciprocity (absorption cross-section, radiation damping, energy-consistent displacement). |
| `acoustic_coupling.py` | Airborne acoustic coupling: $(ka)^n$ pressure reduction, impedance mismatch, and displacement at specified SPL. |
| `mechanical_coupling.py` | Whole-body vibration base-excitation model: frequency-response function and displacement at specified $a_{\mathrm{rms}}$. |
| `oblate_spheroid_ritz.py` | Rayleigh–Ritz analysis on the full oblate spheroid with Legendre-polynomial trial functions and Gauss–Legendre quadrature. |

| Module / Script | Description |
|---|---|
| `multilayer_wall.py` | Classical laminate theory for the multi-layer abdominal wall; effective composite properties and frequency comparison. |
| `viscous_correction.py` | Stokes boundary-layer analysis: viscous damping ratio, Q-factor correction, crossover viscosity, and power-law (non-Newtonian) extension. |
| `organ_inclusions.py` | Hashin–Shtrikman effective medium theory for solid organ inclusions; parametric study of volume fraction and stiffness. |
| `nonlinear_analysis.py` | Donnell–von Kármán cubic stiffness, Duffing backbone curve, jump phenomenon, and amplitude-dependent damping. |
| `parametric_analysis.py` | Full factorial parametric sweep (486 configurations) over $E$, $a$, $c/a$, $h$, $\eta$. |
| `uncertainty_quantification.py` | Monte Carlo and Sobol sensitivity analysis ($N$ = 10,000 samples). |
| `gas_pocket_resonance.py` | Minnaert bubble resonance for intestinal gas pockets. |
| `gas_pocket_detailed.py` | Detailed sub-resonant gas-pocket response at infrasonic frequencies. |
| `orifice_coupling.py` | Upper GI tract (mouth-to-stomach) acoustic coupling pathway. |
| `mechanotransduction.py` | PIEZO1 and PIEZO2 activation thresholds and SPL required for channel opening. |
| `dimensional_analysis.py` | Buckingham-$\pi$ dimensional analysis of the coupled system. |
| *Core library* (`src/browntone/`) | |
| `analytical/` | Shell vibration and acoustic-mode solvers. |
| `fem/` | FEA mesh generation and solver wrappers. |
| `mesh/` | Mesh utilities for parametric geometries. |
| `postprocess/` | Result aggregation and figure generation. |
| `utils/` | Shared constants, unit conversions, I/O helpers. |
| `cli.py` | Command-line interface entry point. |
| *Scripts* (`scripts/`) | |
| `run_modal_analysis.py` | Run the full modal analysis pipeline (all modes, parametric sweep). |
| `run_convergence_study.py` | Mesh and quadrature convergence study for Rayleigh–Ritz solver. |
| `generate_jsv_figures.py` | Generate all publication figures for the JSV paper. |
| *Data* (`data/`) | |
| `materials/default.json` | Material property database (tissue, fluid). |

| Module / Script | Description |
|---|---|
| `results/fea_modal_results.json` | FEA validation: analytical vs. Ritz vs. various BC configurations. |
| `results/uq_results.json` | Full Monte Carlo and Sobol analysis output. |

## 4.3 Reproducing Figures and Tables

All figures and tables in the main paper can be reproduced from the repository as follows.

1. **Install the package:**

   ```
   pip install -e ".[all]"
   ```

2. **Run the full analysis pipeline:**

   ```
   python scripts/run_modal_analysis.py
   ```

   This produces all intermediate results in `data/results/`.

3. **Generate all JSV figures:**

   ```
   python scripts/generate_jsv_figures.py
   ```

   Output appears in `paper/figures/`.

4. **Run individual validation studies:**

   ```
   python -m src.analytical.oblate_spheroid_ritz
   python -m src.analytical.multilayer_wall
   python -m src.analytical.viscous_correction
   python -m src.analytical.organ_inclusions
   python -m src.analytical.nonlinear_analysis
   ```

5. **Run uncertainty quantification:**

   ```
   python -m src.analytical.uncertainty_quantification
   ```

6. **Build the paper:**

   ```
   cd paper && pdflatex main.tex && bibtex main
   pdflatex main.tex && pdflatex main.tex
   ```

7. **Run tests:**

   ```
   make test
   ```

A Docker environment is also provided for fully reproducible builds:

```
docker compose -f docker/docker-compose.yml run --rm browntone bash
```